

Bongo_Seek

Introduction

Bongo_Seek is an AI-powered chatbot that leverages **Deep Seek's 1.5B model** to generate intelligent responses. It is designed to assist users with natural language queries and provide meaningful answers in real time. The application is built using **Python (Flask) for the backend, React and Tailwind CSS for the frontend, Node.js for server-side processing, and MySQL as the database**. Additionally, **Hugging Face** is used to integrate the Deep Seek model.

Features

- **AI Chatbot:** Uses Deep Seek's 1.5B model to generate intelligent responses.
- **User Authentication:**
 - **Signup/Login:** Users can create accounts and log in securely.
 - **2-Factor Authentication (2FA):** OTP-based email verification for enhanced security.
- **Session Management:**
 - Secure JWT-based authentication.
 - Session timeout handling.
- **Search History:** All the prompts and responses are being stored in the search_history table.
- **Secure API Communication:** Ensures safe interactions between the frontend and the backend.

Technologies Used

- **Backend:**
 - Python (Flask)
 - Hugging Face (Deep Seek 1.5B Model)
 - MySQL (Database)

- Flask-JWT-Extended (Authentication)
- Flask-Bcrypt (Password Hashing)
- PyMySQL (Database Connectivity)
- **Frontend:**
 - React.js
 - Tailwind CSS
 - Node.js (for additional processing)
- **Third-party Integrations:**
 - Hugging Face Transformers
 - Two-Step Verification
 - Email (SMTP for OTP verification)

Installation and Setup

Prerequisites

Ensure you have the following installed:

- Python 3.9+
- Node.js & npm
- MySQL Server

Backend Setup

- Clone the repository:
`git clone https://github.com/your-repo/Bongo_Seek.git`
- `cd Bongo_Seek/backend`
- Installed Flask:
`pip install flask`
- Install dependencies:
 1. `pip install transformers`
 2. `pip install Flask-Mail`
 3. `pip install flask cors`
 4. `pip3 install torch torchvision torchaudio --index-url https://download.pytorch.org/whl/cu118`

5.pip install accelerate

- Set up MySQL database:
CREATE DATABASE bongo_seek;
- Run the Flask server:
flask run --port=8080

Frontend Setup

Navigate to the frontend directory:

```
cd ../frontend
```

1. Install dependencies:
npm install
npm install axios
2. Start the React development server:
npm run dev

Usage

1. User Signup & Authentication

- Visit the signup page (</signup>).
- Enter details and submit.
- Verify OTP sent via email.
- After successful verification, navigate to login page.

2. Chatbot Interaction

- Enter a query in the chatbox.
- The AI processes the input and returns a response.

3. Two Factor Verification

- Click **Sign in**.
- Sends OTP code to mail.

API Endpoints

Authentication

Method	Endpoint	Description
POST	/signup	Registers a new user
POST	/login	Authenticates user
POST	/Verify2FA	Verifies OTP
GET	/protected	Authenticated route

Chatbot

Method	Endpoint	Description
POST	/api/chat	Sends a query to the chatbot

Security Measures

- Passwords stored using **bcrypt hashing**.
- JWT authentication for session management.
- **2FA with email OTP** for additional security.
- SQL injection prevention with parameterized queries.

Future Improvements

- Implementing voice interaction.
- Prompt history will show up to five responses in the left side bar.
- Implementation of Google Authentication.
- Forget Password option.
- Bengali language support.
- Enhanced AI model integration.

Contact

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